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8.3.5 Practical Conjugate Gradient Method algorithm

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transpose $A^T r^{(k)} / p^{(k-1)T} A p^{(k-1)}$. Okay?

So now we can take this insight and we can say okay when we're ready to compute the next search direction, all we do is we check whether this is the first iteration in which case we set it equal to the right hand side. Otherwise we compute this γ_k that we just described and then we take our next search direction to be a linear combination

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Homework 8.3.5.1

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```
Given :  $A, b$   
 $x^{(0)} := 0$   
 $r^{(0)} := b$   
 $k := 0$   
while  $r^{(k)} \neq 0$   
  if  $k = 0$   
     $p^{(k)} = r^{(0)}$   
  else  
     $\gamma_k := -p^{(k-1)T} A r^{(k)} / p^{(k-1)T} A p^{(k-1)}$   
     $p^{(k)} := r^{(k)} + \gamma_k p^{(k-1)}$   
  endif  
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$   
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$   
   $r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$   
endwhile
```

we compute

$$\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}.$$

Show that an alternative formula for α_k is given by

$$\alpha_k := \frac{r^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}.$$

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Solution:

For a complete solution see the [notes](#).

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